Not-ruled-out images of policy contrasts

An applied project for ECN 593

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This note writes the pipeline in generality first, then uses one implemented simulator as an example. The reporting object is a *not-ruled-out image* of a named policy contrast. That is the same *reporting logic* as a partially identified set—do not replace a set by a convenient point—not a claim that the object has the theorems of a Manski identified set.

This is the formal write-up. Pre-wave lecture drafts are not part of this packet. Numbers below come from the planted JSON in `results/` and the archived F-empty file. Foil 1 on this pin is OLS-HC1, not a forest and not BCF. Overlay verdicts are written as set relations on subsets of $\mathbb{R}$ (Section 6).

1 The planner's problem

A planner compares assignment rules (policies) that act on a population of interfering units. Individual contrasts $Y_i(1) - Y_i(0)$ are not the target: under interference the argument of Y_i is the whole assignment vector. The natural estimand is a contrast of rules,

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}(\pi')\big]\,, \tag{1}$$

where c indexes a complete data-generating process (a kernel, an ABM, a structural model) and the expectation is over whatever residual randomness the kernel still has once c is fixed.

The planner does not know c. What she has is a class of kernels $\{f(\cdot; c) : c \in C\}$ and a low-dimensional vector of moments z that she is willing to treat as the right functionals of the system. ("Willing" here is an identifying assumption about *which summaries to match*. It is not who named θ. Augusto froze θ_2 as a contrast of terminal means in SPECIFICATION.md §4.) History matching then *rules out* calibrations. It does not produce a posterior over worlds. The decision-relevant object is not the surviving set in parameter space. It is the image of θ over that set.

Two further incompletenesses sit beside calibration, and must not be folded into the same interval: (i) the scoring of the outcome Y (which operationalization?), and (ii) the fact that default estimators are one-world objects. The procedure below keeps the three sources on separate axes.

2 Objects

Kernel. A complete mapping f from (c, ω, π) to a panel (Y_T, D, X, A). Here ω is simulation noise, π is an assignment rule, A is a (possibly stochastic) network, X are covariates.

The kernel is complete: incompleteness lives in c, not in unobserved counterfactuals inside f (Section 3). Implemented as vectorized algebra (`kernel.py`), not as a turtle-by-turtle framework. NetLogo would be a twin of the same f, not a different science.

Box. $C \subset \mathbb{R}^d$ is the a priori range of free coordinates of c. Coordinates that are not up for calibration (in the example, η) are frozen inside f, not inside C.

Named estimands. A short list of contrasts, declared before any wave. Under interference the argument of Y_i is the whole assignment, so

$$\theta(\pi, \pi'; c) \;=\; \mathbb{E}_c\big[\bar{Y}_T(\pi)\big] \;-\; \mathbb{E}_c\big[\bar{Y}_T(\pi')\big]\,. \tag{2}$$

In the illustration the named list is

$$\theta_1(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_1] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0], \tag{3}$$

$$\theta_2(c) = \mathbb{E}_c[\bar{Y}_T \mid \pi_2] - \mathbb{E}_c[\bar{Y}_T \mid \pi_0] \quad \text{(headline)}, \tag{4}$$

$$\psi(c) = \theta_2(c) - \theta_1(c) \quad \text{(targeting premium)}. \tag{5}$$

$\delta_{\mathsf{spill}}(c)$ is a separate exposure-mapped spillover on the only arm that has a design (Section 5.7). Notation: the history-matching threshold is τ_{I}, not ψ. The two must not share a glyph. Expectations are over ω only (π_0 and π_2 are deterministic given A; π_1's extra randomness is part of ω).

θ is defined on the *terminal* village mean $\bar{Y}_T$, not on the 20-day path. That is how the functional was named, not a proof that path summaries would be wrong. A path functional would be a different freeze, and then M and z should be path summaries too.

Moment map. $M : C \to \mathbb{R}^k$ with $k < d$ by design. Underidentification in c is intended; the deliverable is a set in c-space. Whether θ moves on that set is empirical. M is not a menu of outcome scorings.

Anchors. $z \in \mathbb{R}^k$ is the target for M. In a simulation study z may be planted from a known $c^\star \in C$. In an empirical study z is external. Either way, z is frozen before waves.

Implausibility.

$$I(c) \;=\; \big(z - \widehat{M}(c)\big)^\top V(c)^{-1} \big(z - \widehat{M}(c)\big), \tag{6}$$

a squared Mahalanobis discrepancy, not a distance. $V(c) = \mathsf{diag}\big(\sigma^2_{\mathsf{samp}}(z) + \sigma^2_{\mathsf{MC}}(\widehat{M}(c)) + \delta^{\circ 2}\big)$. The three pieces are sampling error in z, Monte Carlo error in the candidate moment, and an explicit model-discrepancy budget δ. V does not contain survey-measure noise.

Not-ruled-out-yet set. $\mathsf{NROY} = \{c \in C : I(c) \le \tau_{\mathsf{I}}\}$. Membership means we are not entitled to discard c. It does not mean c is observationally equivalent to the truth, and it is

not a posterior.

Images.

$$\Theta_{\text{hm}} = \{\theta_2(c) : c \in \text{NROY}\}, \qquad \Theta_{\text{exp}} = \{\delta_{\text{spill}}(c) : c \in \text{NROY}\}. \tag{7}$$

Reported as empirical $[L, U]$ over a pre-registered sample of survivors. No measure (uniform, posterior, …) is placed on NROY. Width is empirical: distinct survivors *can* map to different θ; they need not.

Diagnostic worlds. Two points $c_0^{\min I}$, $c_0^{\text{med}\, I}$, functions of I only. Never of θ. "Best match" and "non-extremal" are the rationales; neither is theoretically privileged.

Measures. Scorings m of Y_T, frozen as a menu, admitted or excluded by a pre-declared restriction. $M^\star$ is a set of measures, not a subset of NROY. $\Theta_{\text{cv}}(c_0)$ is the image of the estimand over $M^\star$ at that world.

Panel. One run of f is one micro-dataset

$$\mathcal{D}(c, \omega, \pi) \;=\; \big\{(Y_{iT},\, D_i,\, X_i,\, A_{i\cdot})\big\}_{i=1}^{N}. \tag{8}$$

World $\neq$ panel. One-world means the estimator is not given NROY; it conditions on one c. One panel is one ω. Kernel truth $\theta(c)$ is still one-world and uses $R = 200$ paired reps. Averaging a foil over 200 panels at the same c_0 would still be one-world.

Foils. One-world estimators fitted at each c_0 on *one* panel. Each foil is judged against *its own* image and against truth at that world (Section 6). They are not candidates for Θ_{hm}.

3 Potential outcomes, and what counts as data

Hahn / Rubin–Neyman with SUTVA: $Y_i = Y_i(D_i)$, $D_i \in \{0, 1\}$. One of two numbers per person is missing. That missingness *is* the incompleteness. BCF, an honest forest, or OLS-HC1 is a gold-standard interval for that SUTVA functional, given one dataset, at one world. The ruler, not the villain. Not "the forest failed under interference."

This class drops SUTVA ($\gamma > 0$, peer *outcomes*): $Y_i = Y_i(\mathbf{d})$, $\mathbf{d} \in \{0, 1\}^N$. The individual contrast $Y_i(1) - Y_i(0)$ is undefined unless everyone else's assignment is named. The well-defined object is a Hudgens–Halloran overall / policy contrast $\theta(\pi, \pi'; c)$ as above.

In the *field*, other $\bar{Y}(\pi)$ are still missing (one campaign). That would be Manski incompleteness in the potential outcomes. In the *simulator*, f is complete: run both arms, pair ω, subtract. Nothing inside f is an unobserved counterfactual. Remaining incompleteness is which $c \in C$ (five free knobs, three moments). $\Theta_{\text{hm}} = \{\theta_2(c) : I(c) \le \tau_{\text{I}}\}$

is a not-ruled-out image, not a Manski identified set and not a set of still-possible $Y_i(0)$. Same *reporting logic* (do not replace a set by a convenient point); not the same theorem.

$d > k$ underidentifies c. Width of Θ_{hm} is measured. Matching arm *levels* while θ_2 is a *contrast against control* makes unpinning possible, not a theorem (Section 5.5).

Only two objects have empirical analogs.

empirical analog	simulator object	who consumes it
one campaign's micro data	one panel $\mathcal{D}(c, \omega, \pi)$	OLS-HC1, Hájek
calibration targets	z (plus sampling variances, δ)	$I(c)$, NROY
survey operationalization	$s = m(Y_T)$	foils / Θ_{cv}

Not data: LHS points, moment and estimand Monte Carlo reps, NROY survivors, the image sample, $\theta(c)$. Those are probes of f. NROY points are arguments of $\theta(\cdot)$, not extra surveys.

4 Order of operations

The order is part of the identification statement. Using θ to decide which c survive is fishing.

1. Freeze f, C, z, δ, τ_{I}, the estimands, the measure menu and its admission rule, and the c_0 rule.
2. Draw the environment (A, X) once.
3. History-match: evaluate $I(c)$ on a space-filling design, keep $I \leq \tau_{\text{I}}$, optionally concentrate later waves on the surviving box. NROY is the union of survivors. Empty NROY is a finding: the pair (f, z) is rejected. Do not move τ_{I} or δ after seeing the waves.
4. Leg A. Sample survivors. Compute $\theta(c)$ and $\delta_{\text{spill}}(c)$ from the kernel. Report Θ_{hm} and Θ_{exp} as $[L, U]$.
5. Leg B. At each c_0, fit the one-world foils on a single panel. Compare each interval to its own image and to kernel truth at that c_0.
6. Measures. At those same c_0, score Y under each m, admit, report Θ_{cv}. Side branch, not a gate on legs A and B.

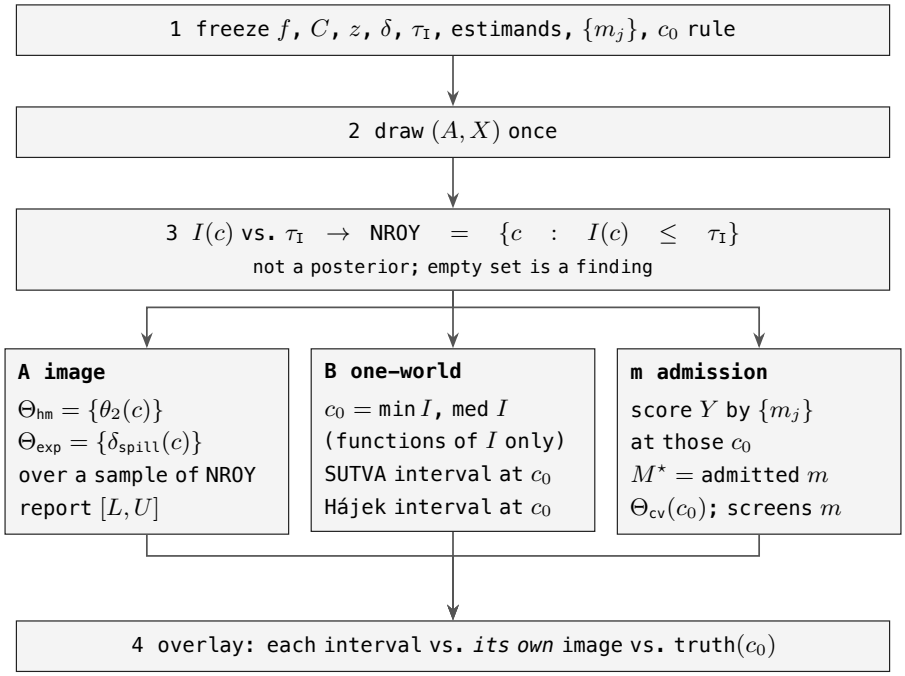


Figure 1: Order of operations. Legs A and B, and the measurement branch, are three reports. They do not merge into a single "construct-identified-robust" object.

5 Design of exogenous quantities

Everything below is frozen before waves. The justifications are design justifications, not data-dependent ones.

5.1 Threshold τ_{I}. $I(c)$ is a k-dimensional squared Gaussian form if $z - \widehat{M}$ were normal with covariance V. Taking $\tau_{\mathrm{I}} = F^{-1}_{\chi^2_k}(0.95)$ is the conventional history-matching cutoff (a 95% ellipsoid). It is a ruling-out convention, not a posterior credible region. Moving τ_{I} after seeing NROY would convert a pre-registered filter into a fit.

5.2 Discrepancy δ. Kennedy–O'Hagan discrepancy is the admission that even the right c need not hit z exactly, because f is not the world. On a correctly specified simulator δ is conservative: NROY is larger than a $\delta = 0$ match. Keeping a pre-declared δ anyway separates "the analyst's budget" from "what the DGP can hit." Loosening δ after a wave is the same sin as loosening τ_{I}.

5.3 The box C. Wide enough that underidentification is possible ($d > k$), narrow enough that every coordinate is economically signed (persistence, peer weight, and own-treatment shift positive; intercept not so free that Y saturates for every c). Excluding $\gamma = 0$ is a statement of subject-matter: the paper is about interference; a no-interference world surviving calibration would be its own finding and must not be slipped in.

5.4 How c is drawn. Nobody picks a favourite five-tuple by looking at θ. Wave 1 is a Latin Hypercube of 1000 points in C. Each candidate is kept iff $I(c) \leq \tau_{\mathrm{I}}$. Waves 2–3

redraw LHS in the survivors' bounding box, inflated 20% and clipped back to C. NROY is the *union* across waves. The plant $c^\star$ is the centre of C, chosen without looking at θ. Diagnostic c_0 are min I and median I on that union.

5.5 Anchors z. Two admissible sources, with different claims.

External z. Moments with a policy story and a kernel analog. If no $c \in C$ satisfies $I(c) \le \tau_{\mathrm{I}}$, that is F-empty: the pair (f, z) is rejected. That finding is about the analog, not about θ. Empty is the opposite of a wide Θ.

Planted z. $z := \widehat{M}(c^\star)$ at a $c^\star$ chosen without looking at θ. The illustration uses the centre of C. Planting makes NROY feasible by construction and makes θ_{true} known, which is the only place an overlay is checkable.

There are *three* moments, not five. Five is d, the free coordinates of C ($\alpha, \beta, \gamma, \kappa, \sigma_\varepsilon$). η is frozen inside f, not a moment and not in C. The three slots, in order:

slot (historical name)	village functional	arm
level_imm	$\mathbb{E}[\bar{Y}_T \mid \pi_2]$	hub level
gradient_edu	$\mathrm{corr}(Y_T, x_1 \mid \pi_2)$	literacy gradient
level_internet	$\mathbb{E}[\bar{Y}_T \mid \pi_1]$	random level

Control's mean $\mathbb{E}[\bar{Y}_T \mid \pi_0]$ is *not* in z. That is how z was named, not a theorem that it must be omitted. World Bank slots were three campaign-era / country aggregates (immunization, education gradient, internet); there is no untreated planet in that file. After F-empty, AMENDMENT-02 kept the same three functionals and planted them at $c^\star$. SPECIFICATION.md does not contain a paragraph "exclude π_0 so that θ_2 is set-identified." Underidentification of c is designed ($d = 5$, $k = 3$). Unpinning of θ_2 is a *possible* consequence of matching levels while θ_2 is a contrast against an unmatched control, plus δ. Width remains measured.

If control were added, $k = 4$ and c would still be a set, but θ_2 would sit close to a difference of two matched moments plus the δ slop. That would be a different freeze. Neither adding it to manufacture tightness nor dropping it to manufacture width is a scientific reason.

The illustration ran both sources. Country aggregates as z returned F-empty and were archived. The overlay uses the plant.

5.6 Diagnostic c_0. Min I mimics best-fit / MSM practice. Median I is a non-extremal rule on the same ranking. Both are functions of moment fit, not of the estimand. Two worlds, not four hundred: looping the estimator over every survivor answers "what if we gave the learner the calibration set," which default practice does not.

5.7 Foils. Foil 1 is a SUTVA one-world interval at c_0. The frozen role is that ruler: an honest interval *conditional on a world*, not a defendant charged with failing under interference. Preferred software is an honest causal forest; this pin does not load it (econml against numpy==2.4.6). The run therefore uses the declared fallback: OLS of

the scored outcome on D given X, HC1 interval. Report the name. Do not call it BCF. BCF remains an extra row only if asked for, same c_0, no new rule.

On one π_2 panel the regression is $s_i = \alpha + \tau D_i + X_i^\top \eta + u_i$. Stack $Z_i = (1, D_i, X_i') \in \mathbb{R}^k$ with $k = 5$ here (intercept, D, three covariates). OLS $\hat\beta = (Z^\top Z)^{-1} Z^\top s$, and $\hat\tau$ is the D coordinate. HC1 is the degrees-of-freedom sandwich

$$\widehat{\mathsf{Var}}_{\mathsf{HC1}}(\hat\beta) \;=\; \frac{n}{n-k}\,(Z^\top Z)^{-1}\Big(\sum_{i=1}^{n} \hat u_i^2\, Z_i Z_i^\top\Big)\,(Z^\top Z)^{-1}. \tag{9}$$

I_{OLS} is $\hat\tau \pm z_{1-\alpha/2}\,\mathsf{SE}(\hat\tau)$ with SE the square root of the D diagonal. At $N = 800$, $n/(n-k) = 800/795 \approx 1.006$; the name is the convention, not a material correction. Rows are treated as uncorrelated. That is already a SUTVA-style assumption, not a network standard error. On the hub panel D *is* "being a hub," so $\hat\tau$ lives near δ_{dir} (hubs vs non-hubs inside one campaign), not near θ_2.

Foil 2 is design-based and exposure-mapped, and therefore lives only on an arm with a known assignment *lottery*. "Design" here is not "research design" and not "we designed an ABM." It means: can one write $p_i = \Pr_\pi(D_i = 1)$ and exposure probabilities from a known hat, before seeing Y?

π_1 is SRSWOR: simple random sample without replacement. Seed exactly 80 of 800 households; every 80-subset equally likely; no repeats. Inclusion $p_i = 80/800 = 1/10$. Neighbour counts among the untreated are hypergeometric, which is how $p_{i,S}$ are written without touching (α, β, γ). π_2 (top-80 degree) has no hat: given A, $p_i \in \{0, 1\}$. Inverse-probability weights are undefined. That un-reachability is reported, not walked around.

"Estimable from one designed panel" means: from one realized randomized campaign plus the known lottery, form $\widehat{\delta}_{\mathsf{spill}}$ and $\widehat{\delta}_{\mathsf{dir}}$. It does *not* mean θ_2 is identified from one campaign. θ_2 needs π_2 and π_0. Even a field hub-campaign does not see control, and was not a lottery.

Exposure $g_i = (D_i, s_i)$ with the same k_i as in the kernel (degree, isolates patched):

$$k_i = \mathsf{max}\Big(\sum_j A_{ij},\, 1\Big), \qquad s_i = \frac{\sum_{j:A_{ij}=1} D_j}{k_i}. \tag{10}$$

s_i is the share of neighbours' *treatment*, not neighbours' outcomes. Isolates have $k_i = 1$ and $s_i = 0$. Path $A{-}B{-}C{-}D$, treat $\{B\}$: $s = (1, 0, 0.5, 0)$. Strata at cut 0.25 are the sets

$$\begin{aligned} S_1 &= \{i : D_i = 1\}, \\ S_{0a} &= \{i : D_i = 0,\ s_i = 0\}, \\ S_{0b} &= \{i : D_i = 0,\ 0 < s_i < 0.25\}, \\ S_{0c} &= \{i : D_i = 0,\ s_i \geq 0.25\}. \end{aligned}$$

On that path: $A, C \in S_{0c}$, $B \in S_1$, $D \in S_{0a}$, $S_{0b} = \emptyset$. If any cell has fewer than 20 agents in a replication, S_{0b} is merged into S_{0c} for all reps of that world.

Kernel truth uses unweighted cell means of oracle Y, then averages over random-arm reps:

$$\delta_{\texttt{dir}}(c) = \mu_{S_1} - \mu_{S_{0a}}, \qquad \delta_{\texttt{spill}}(c) = \mu_{S_{0c}} - \mu_{S_{0a}}. \tag{11}$$

Foil 2 replaces μ_S by the Hájek mean on one π_1 panel,

$$\widehat{\mu}_S \;=\; \frac{\sum_i \mathbf{1}\{i \in S\}\, Y_i / p_{i,S}}{\sum_i \mathbf{1}\{i \in S\}/p_{i,S}}. \tag{12}$$

Glue the two μ_S and the oracle gate is unreadable. The exposure map is neighbour-D; the kernel peer term is neighbour-Y over T ticks (a Sävje-style misspecification of the map). That is a limitation of Foil 2, not a license to replace it with a second SUTVA learner.

Foil 2 does *not* estimate θ_2. Two different one-world referents, neither of which is the calibrated-simulator report: Hahn classroom $\rightarrow$ Foil 1; a network RCT that admits interference and uses an exposure map (Aronow–Samii family) $\rightarrow$ Foil 2. The goal of the second row is an interference-aware interval that (i) can write spillover, (ii) is identified by a lottery from one panel, (iii) still does not consume NROY, so I_{E2} versus Θ_{exp} isolates calibration after architecture (Section 6). Rank replacements by that goal. Dropping Foil 2, or swapping it for BCF, returns the architecture/calibration confound. Stochasticizing hub-seeding so IPW can hit it is a different policy. Θ_{exp} is the image of the *kernel* $\delta_{\texttt{spill}}(c)$, not of the Hájek interval.

5.8 Measures. A short menu, including a positive control (oracle $s = Y$) and a scoring designed to fail a restriction stated in English before the wave ("must not load on a treatment-channel covariate"). Admission is mechanical. It does not look at θ.

5.9 Monte Carlo budgets. Moment evaluations use fewer replications than estimand evaluations: ruling-out has to be cheap enough to search C; $\theta(c)$ is computed only on survivors. The plant uses the estimand budget so that z is not noisier than the object we will later quote.

6 Interval geometry

Sampling intervals and images are subsets of $\mathbb{R}$. Verdicts are inclusions, not slogans.

Let $I = [\iota_L, \iota_U] \subset \mathbb{R}$ be a closed sampling interval (a foil CI at one c_0) and let $\Theta = [L, U] \subset \mathbb{R}$ be the reported image of the *same* functional over a pre-registered sample of NROY (empirical min/max; not a continuum and not a posterior). Write $\text{int}(\Theta) = (L, U)$.

Implemented labels (SPECIFICATION.md §10; classify_interval):

$$\texttt{F-HONEST} \iff \Theta \subseteq I, \tag{13}$$

$$\texttt{F-OVERCONF} \iff I \subset \text{int}(\Theta). \tag{14}$$

Anything else is unnamed — no third finding after the freeze — including

$$I \cap \Theta = \emptyset \quad \text{and} \quad I \cap \Theta \neq \emptyset \text{ with neither inclusion.} \tag{15}$$

The code does not additionally test whether θ_{true} sits outside I. SPECIFICATION.md's prose "and excludes θ_{true} or parts of $[L, U]$" is already implied by $I \subset \text{int}(\Theta)$ for the image endpoints; $\theta_{\text{true}}(c_0)$ is checked separately by the oracle gate.

Oracle gate (harness, not a finding):

$$\theta_{\text{true}}(c_0) \in I \quad \text{or, for Foil 2,} \quad \delta_{\text{spill}}(c_0) \in I_{E2}. \tag{16}$$

The point estimate is unused. Fail $\Rightarrow$ stop; do not read overlay stickers.

Each foil versus *its own* image:

$$(I_{\text{OLS}},\ \Theta_{\text{hm}}), \qquad (I_{E2},\ \Theta_{\text{exp}}). \tag{17}$$

Cross-functional geometry (I_{OLS} against Θ_{hm} when OLS is not θ_2) is reported as a set relation, not forced into F-$*$.

Two gaps must not be dumped into one interval. Write

$$\mathcal{A} = \text{architecture (SUTVA vs exposure map vs kernel)},$$
$$\mathcal{C} = \text{calibration (which } c \in \text{NROY)}.$$

I_{OLS} versus Θ_{hm} can mix $\mathcal{A}$ and $\mathcal{C}$ when the functionals differ. Foil 2 is the attempt to handle $\mathcal{A}$ first: it can write spillover and still does not see NROY. Then, in order,

1. $\delta_{\text{spill}}(c_0) \in I_{E2}$ — architecture and sampling OK *at the world*;
2. $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$ — residual width is variation of the *same* functional across surviving c.

That pair is what "I_{E2} versus Θ_{exp} isolates calibration after architecture" means. The leftover set $\Theta_{\text{exp}} \setminus I_{E2}$ is not a sampling CI and is not a SUTVA misspecification residual. F-OVERCONF-E2 is only the sticker for (2). It is almost automatic once (1) holds and the length of Θ_{exp} exceeds sampling noise. Empty or covering images are also allowed findings; the interesting object is the width, not the label.

7 Illustration: a planted village

One large village, $N = 800$ households, acquaintance network $G(N, 4/N)$ (mean degree ≈ 4). $Y_{i,t} \in (0, 1)$ is preventive-practice intensity—a score, not a country coverage rate. Covariates: literacy x_1, resources x_2, baseline propensity x_3. Three rules seed 80 households from $t = 0$: none (π_0); SRSWOR (π_1, the only arm with a design); the 80 highest-degree households (π_2, deterministic given A).

The kernel is linear-in-means with logistic saturation. For $t = 0, \ldots, T-1$ with $T = 20$,

$$Y_i(t+1) \;=\; \Lambda\Big(\alpha + \beta Y_i(t) + \gamma \frac{\sum_{j \in N(i)} Y_j(t)}{k_i} + \eta \cdot X_i + \kappa D_i + \varepsilon_i(t)\Big), \tag{18}$$

$\Lambda(z) = 1/(1+e^{-z})$, $\varepsilon_i(t) \sim N(0, \sigma_\varepsilon^2)$. Peers enter as neighbours' *lagged practice*, not as neighbour treatment. The five free coordinates and the frozen η are

	box / value	role
α	$[-2.5, -0.5]$	intercept of the index
β	$[0.05, 0.60]$	own-lag persistence
γ	$[0.05, 0.80]$	peer weight; 0 excluded
κ	$[0.25, 2.50]$	own-treatment shift (same κ on every arm)
σ_ε	$[0.01, 0.40]$	sd of ε, not a standard error
η	$(0.05, 0.10, 0.02)$	literacy, resources, propensity; not calibrated

$c^\star$ is the centre of C: $(-1.5,\ 0.325,\ 0.425,\ 1.375,\ 0.205)$. Frozen $z = \widehat{M}(c^\star) = (0.264,\ 0.050,\ 0.262)$ – hub-arm mean Y_T, corr(Y_T, x_1) on the hub arm, random-arm mean Y_T. The control mean is *not* in z, so θ_2 (a contrast against control) is not a moment being matched. $\delta = (0.02, 0.05, 0.02)$ and $\tau_{\mathrm{I}} = F^{-1}_{\chi^2_3}(0.95) \approx 7.81$ are the same numbers used in the rejected country-total attempt. The graph A is drawn once and not calibrated. Connectivity is not the latent being measured. The measurement layer scores Y, not edges.

8 PoC results

8.1 External z failed. Matching the same kernel to rescaled World Bank country totals (DPT coverage, education–immunization correlation, internet users; fetched 2026-08-12) ruled out every c. Wave 1 kept **0** of **1000**. Minimum $I = 204$ against $\tau_{\mathrm{I}} \approx 7.81$. Closest M in the box is still far from z (Figure 2). τ_{I} and δ were not moved. That finding stands for the pair (kernel, country aggregates). It is not a wide Θ. It is a reason to stop asking this kernel to match functionals it cannot speak to.

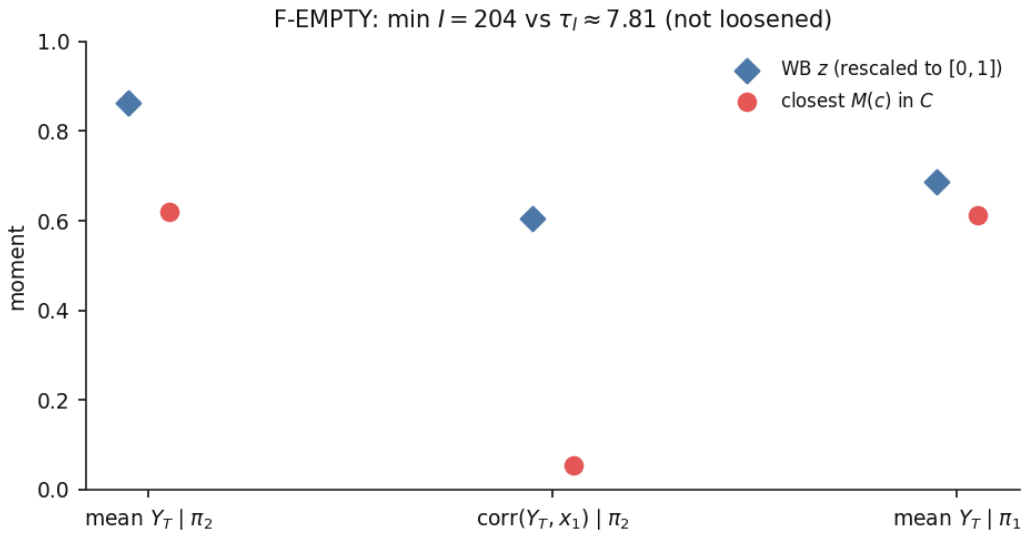


Figure 2: F-EMPTY on World Bank z (rescaled to $[0,1]$). Closest simulated moments in C versus anchors. Not a calibrated world.

8.2 Planted z left a fat NROY. Against $z := \widehat{M}(c^\star)$, three waves kept 202, then 416, then 373; union $|\text{NROY}| = 991$ (Figure 3). The set did not collapse. Closest survivor sits next to $c^\star$ ($I \approx 0.002$), which is what a correctly specified plant should do. That point is $c_0^{\min I}$ by the pre-declared rule, not because it won on θ. Median-I c_0 has $I \approx 2.02$ and a much larger κ.

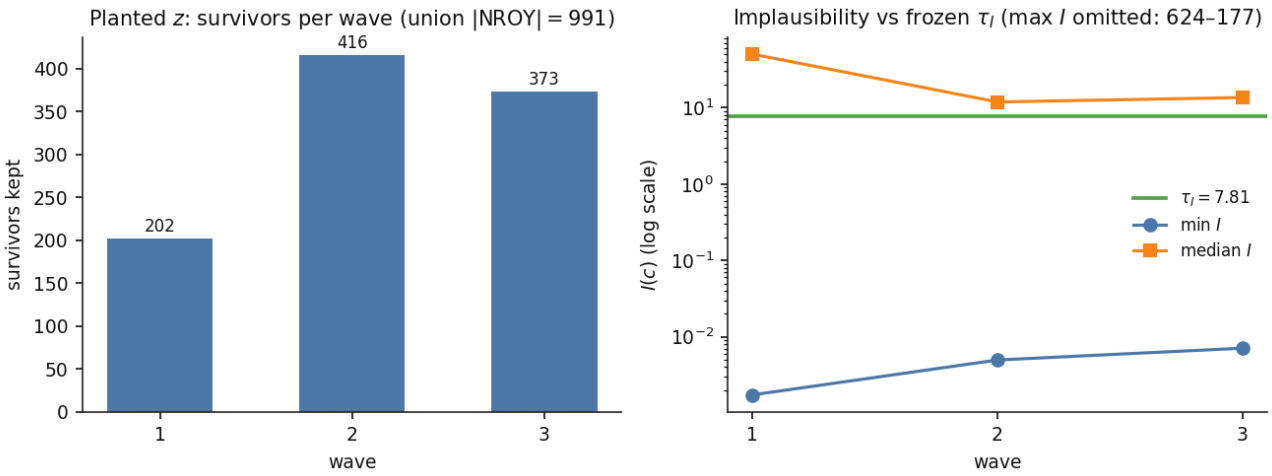


Figure 3: Planted history matching. Left: survivors per wave. Right: min and median $I(c)$ against frozen τ_I (log scale). Later waves concentrate; τ_I is not moved.

8.3 The image of θ_2 is not a point. At $c^\star$, hub vs none is $\theta_2 = 0.038$ (3.8 pp). Random vs none is $\theta_1 = 0.036$. Targeting premium $\psi \approx 0.002$. In this DGP, picking influencers barely beats picking at random. That is a fact about this $c^\star$, not a theorem about targeting.

Among 400 sampled survivors, θ_2 runs from 0.6 pp to 7.3 pp. The planted 3.8 sits inside. Honest report: $\Theta_{\text{hm}} = [0.006, 0.073]$, not 0.038. Not a sampling distribution and not a posterior.

Spillover $\delta_{\text{spill}}(c^\star) \approx 0.008$; image $\Theta_{\text{exp}} \approx [-0.001, 0.027]$. Direct-effect analogue (hubs vs non-hubs *inside* the random campaign) $\delta_{\text{dir}}(c^\star) \approx 0.34$.

object	min I world	median I world
$I(c_0)$	0.002	2.02
$\theta_2(c_0)$	0.037	0.063
OLS-HC1 I_{OLS} (hub panel)	$[0.326, 0.346]$	$[0.583, 0.595]$
I_{OLS} vs Θ_{hm}	$I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$	$I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$
$\delta_{\text{spill}}(c_0)$	0.006	0.008
Hájek I_{E2}	$[0.003, 0.009]$	$[0.006, 0.011]$
I_{E2} vs Θ_{exp}	$I_{E2} \subset \text{int}(\Theta_{\text{exp}})$	$I_{E2} \subset \text{int}(\Theta_{\text{exp}})$
oracle gate	$\delta_{\text{spill}}(c_0) \in I_{E2}$	$\delta_{\text{spill}}(c_0) \in I_{E2}$
Θ_{cv}	$[0.030, 0.037]$	$[0.056, 0.063]$
admitted m	m_1, m_2	m_1, m_2

8.4 Overlay. Figure 4 is the money plot (two c_0, two functionals). Figure 5 repeats the same numbers on a shared axis so the functional mismatch is visible without faceting.

At both c_0, $\delta_{\text{spill}}(c_0) \in I_{E2}$ (oracle gate: harness not broken) and $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$. Architecture handled; residual is which surviving c. The sticker F-OVERCONF-E2 is that second inclusion.

OLS-HC1 on the hub panel lives near 0.33–0.59, the neighbourhood of δ_{dir}, not of θ_2. D *is* "being a hub." The geometry is $I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$, which is unnamed (neither $\Theta_{\text{hm}} \subseteq I_{\text{OLS}}$ nor $I_{\text{OLS}} \subset \text{int}(\Theta_{\text{hm}})$). Not "OLS failed under interference."

8.5 Declared cut at 0.5. SPECIFICATION.md already required this check: recompute the exposure strata at threshold 0.5, and report whether the E2 sticker flips. It was run on 24 September 2026. τ_{I} and δ were not touched. The cut changes the functional, so the interval is judged against a rebuilt image, not against the headline Θ_{exp}.

History matching was re-run and reproduced the freeze: waves kept 202, 416, 373, $|\text{NROY}| = 991$, and the same two c_0. On a fresh draw of 400 survivors, the same sample rule as the headline image, δ_{spill} at cut 0.5 has image approximately $[-0.00003, 0.015]$.

object at cut 0.5	min I world	median I world
$\delta_{\text{spill}}(c_0)$	0.0037	0.0047
Hájek I_{E2}	$[0.0019, 0.0069]$	$[0.0033, 0.0077]$
I_{E2} vs the 0.5 image	$I_{E2} \subset \text{int}(\Theta)$	$I_{E2} \subset \text{int}(\Theta)$
oracle gate	truth inside I_{E2}	truth inside I_{E2}
S_{0b} merged	yes	yes

The sticker does not flip. At both c_0 the interval is still a strict subset of the

interior of its own image, and the oracle truth at the new cut sits inside the interval. The code name remains F-OVERCONF.

The cut is not innocuous. In all 400 sampled worlds, and at both diagnostic worlds, S_{0b} had fewer than 20 agents, so the pre-registered merge of S_{0b} into S_{0c} fired. The no-flip result is a statement about that merged functional. It is not evidence that the strata were unchanged. Numbers are in `results/appendix_cut_0.50.json`.

Oracle and noisy-survey scorings of Y are admitted; the item that loads on baseline propensity is excluded, as designed. Θ_{cv} is a small scoring gap at a world, not a second NROY.

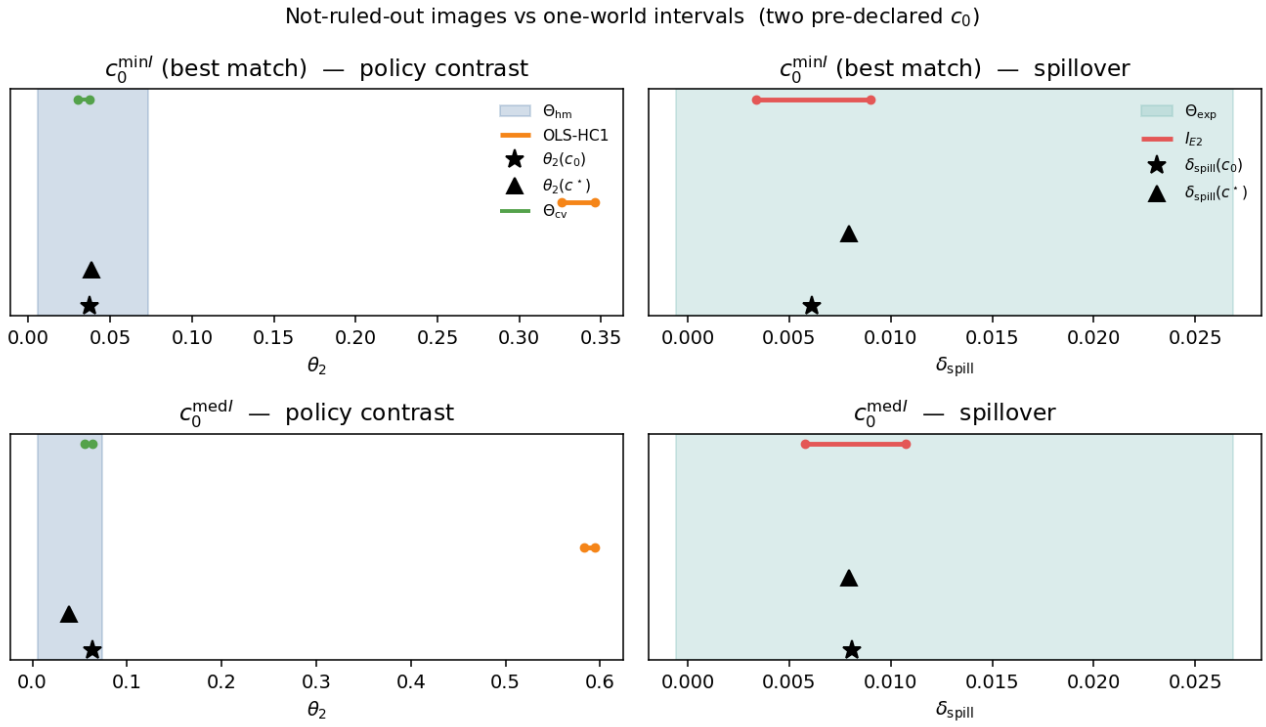


Figure 4: Not-ruled-out images versus one-world intervals at two pre-declared c_0. Left: θ_2 against Θ_{hm}; orange is OLS-HC1 (not a forest). Right: δ_{spill} against Θ_{exp}; red is Hájek. Each interval against *its own* image.

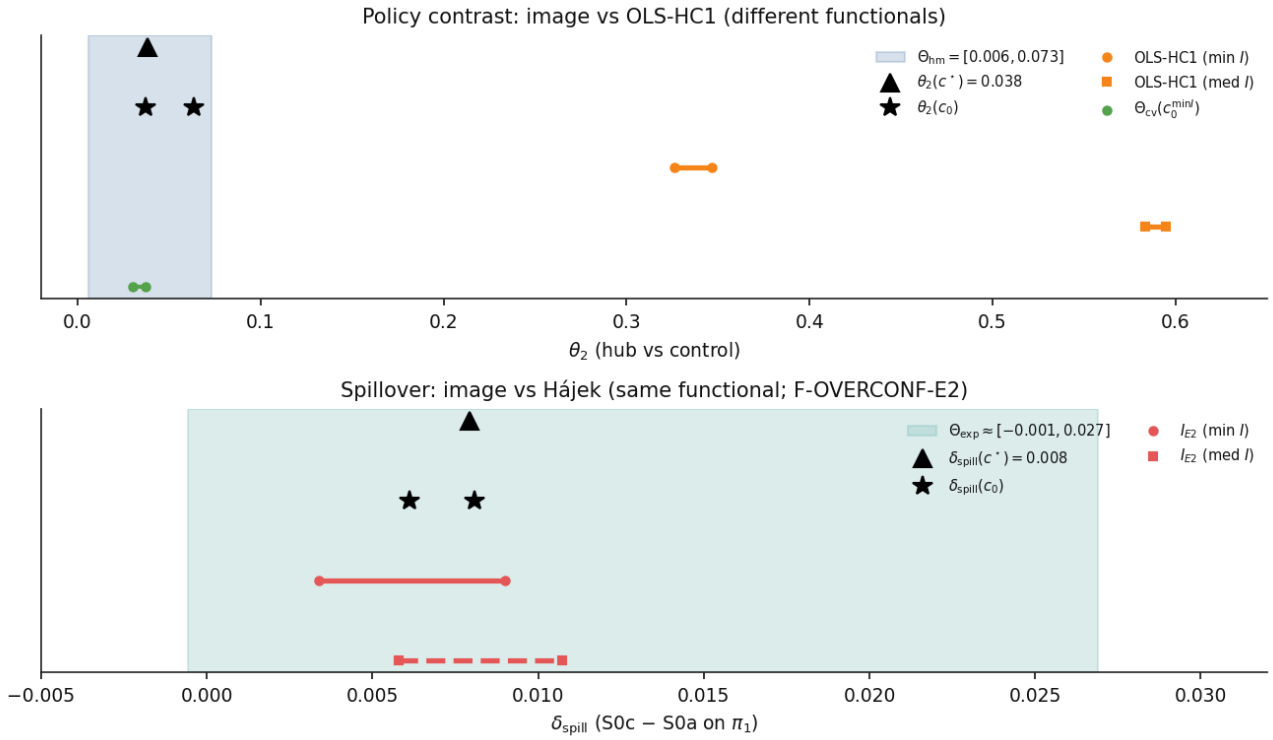


Figure 5: Same objects on one axis per functional. Top: $I_{\text{OLS}} \cap \Theta_{\text{hm}} = \emptyset$ because OLS is not θ_2. Bottom: $\delta_{\text{spill}}(c_0) \in I_{E2}$ and $I_{E2} \subset \text{int}(\Theta_{\text{exp}})$.

9 Who should care — and who should not

Slogan-level novelty is weak. History matching already produces NROY in input space (Craig, Goldstein, Vernon). Equifinality of ABMs is not a discovery of this note (Beven; Williams–Guikema–Brown–Agrawal map ensembles through policy). Partial identification already says: report the set. Interference already has overall vs direct (Halloran, Hudgens–Halloran). Honest forests and BCF already give intervals *given a panel*. None of those pieces is new. The remaining wedge, if this is ever a paper, is the *composition*: a named policy contrast under interference, as the image of a history-matched set, plus a measure-admission set, plus a paired SUTVA ruler and an interference-aware estimator that isolates calibration as the residual, on a simulator where truth is known. That is enough for an M.S. demonstration. It is not an Econometrica gap.

Anyone (method). Interesting if you already publish a policy number from a *calibrated simulator* (ABM, structural model, microsimulation) and currently report Monte Carlo error at a staff c_0. The overlay is a reporting split: sampling at a world is not uncertainty about which worlds z still allows. Empty NROY is a usable negative result (the analog is rejected; do not loosen the cutoff). Width of Θ is measured, not assumed: here θ_2 moves by about 7 pp across survivors, not by "anything."

Causal inference, empirical. Not a new identifying strategy for the designs STP 517 actually teaches. This procedure does not replace unconfoundedness, an instrument, DiD, or an RDD. It does not estimate a CATE $\tau(x)$ on a network. History matching does not identify heterogeneous treatment effects.

It *is* relevant to empirical CI in two narrower cases.

(1) *You are already using a model as the identification vehicle* — a structural DGP, an epidemic simulator, an ABM of a campaign — and then quoting one $\hat{\theta}$. Then the honest object is the image of that θ over calibrations not ruled out by the moments you were willing to match. One-world intervals (regression, forests, BCF) remain honest *at a world*. They are not the image. If your paper is IV for wages, this is not your appendix.

(2) *Interference is in the DGP you wrote, and ITE is undefined.* The well-posed object is a contrast of assignment rules. A SUTVA learner on a targeting arm, where D is a function of the graph, identifies something in the neighbourhood of "being a hub," not the campaign contrast. That is an identification remark, not an estimator bake-off. It is closer to STP 517 (name the estimand; when is regression causal) than to "run BCF on our panel."

This PoC, specifically. The overlay is checkable because z was *planted*. The one attempt at external z (World Bank country totals) returned F-empty. So this is not an empirical estimate of immunization policy, and it should not be narrated as one. As a demonstration that the composition can be run, with pre-registered verdicts including F-empty and "neither F-$*$", it is a completed lab. As a contribution to empirical causal inference on real networks or real campaigns, it is not more than that lab — a toy with a clean object dictionary — unless someone later freezes a z this kernel can actually speak to, without moving τ_{I}.

If the audience is "applied micro who run regressions," treating this as a portfolio piece that shows you can name functionals and not cheat a cutoff is the honest pitch. If the audience already calibrates simulators for policy, the overlay is the thing to steal; the village story is not.

10 What the procedure is not

- Not a posterior over c, and not a distribution of treatment effects.
- Not a Manski identified set theorem-for-theorem. Same reporting logic; different incompleteness (calibration of a complete kernel, not missing counterfactuals); sampling and discrepancy already folded into V.
- Not a claim that equifinality of θ is guaranteed. Width is measured, not assumed.
- Not a finding that one-world estimators are bad at networks. They are honest for a world. The overlay asks whether one calibrated world is enough.
- Not a measurement model of the network. A is frozen.
- Not an empirical immunization study. WB z was rejected.

Implementation. ecn593-poc-abm, AMENDMENT-02. $|\text{NROY}| = 991$, $\Theta_{\text{hm}} = [0.006, 0.073]$, $\theta_2(c^\star) = 0.038$. Foil 1 estimator: OLS-HC1 (CausalForestDML did not import; not BCF). Overlay: figures/overlay.png. WB attempt archived as F-empty. Headline figures are redrawn from the planted JSON. The 0.5 cut was recomputed on 24 September 2026; history matching reproduced the freeze.